1. The steps, from paper to decision

What happens when a paper is put into Lightyear, in order. A person decides twice: at Gate A and at Gate B.

The model reads and writes · Code calculates · A person decides · What you get

BEFORE ANY BACKTEST (NO MARKET DATA NEEDED)

00 Triage
Is this a paper worth testing at all?

→

01 Ingest
Read the paper. Work out the strategy and which Indian universe to test it on.

→

02 Strategy card
Write the rules as one card: signal, weights, rebalance, cost, cash rate, what counts as failure.

→

Data map + analyst read
What data the rules need. The model's one-page view of the paper, written now, before any result exists.

↓

GATE A: you approve the card
You read the card and the analyst read, ask for changes if needed, and approve. The card and the read are locked together with a hash at that moment, so neither can be edited after results exist. The cost and cash rate are whatever the card says.

↓

THE BACKTEST

03 Data check
Are the prices, index members and index levels on file for these rules?

→

04 Point-in-time data
Build the data as it was known on each date. Delisted stocks stay in.

→

05 Run
The run's script trades the card's rules day by day: signal at one close, fill at the next.

→

06 Research checks
Bootstrap, sub-periods, cost and rule sensitivities.

→

07 Portfolio checks
Factor regression, simpler alternatives, capacity.

↓ the run hands over its files: daily returns, holdings, trades, comparators, checks

WHAT LIGHTYEAR BUILDS FROM THOSE FILES (NO NEW BACKTEST; "REBUILD" REPEATS ONLY THIS PART)

Rule check
Re-reads the run's files against the approved card: did the backtest do what the card says?

Monthly figures
Complete calendar months only. Returns, risk ratios, market phases, rolling returns, factor attribution, tail risk, capture, allocation.

Daily figures
The same strategy on daily returns, kept beside the monthly ones.

Gate B score (v4)
Three hard blocks and three scorecard rows, at the card's cost.

↓

Results tab
Tiles, financial-year returns, NAV chart, trailing returns, market phases, rolling returns, factor attribution, drawdown, tail risk, capture, calendar years, allocation. Daily calculations and run diagnostics are folded below.

Main workbook
The same sections with live Excel formulas. A "Python vs Excel" sheet checks every figure to 1e-9.

Diagnostics workbook
The working: every holding and trade, alternatives, cost sweeps, data checks, Gate B rows.

Analyst review
The model's plain-English read of the results. An opinion, not a decision.

↓

GATE B: you decide
Hard blocks (all must pass): data integrity · edge over the NIFTY 500 at the card's cost · paired bootstrap (could the better Sharpe be luck?).
Scorecard (informs, does not block): sub-period stability · factor alpha · performance after publication.
The page shows PASS or FAIL. The decision stays PENDING until you record APPROVE, OBSERVE, FIX or REJECT with your name and reason.

→

08 Record
The decision is written to the run folder and the decisions ledger. Nothing is traded by Lightyear.

The one rule: the model reads and writes, code calculates every number, a person makes every decision. Experimental Gate B tests are kept on a separate branch and do not count until you choose them.

2. Where the data comes from

Every price, index member and index level comes from three files. Nothing is downloaded during a run.

A file someone supplied · Built by code from those files · A setting a person chose · Not on file

SOURCE FILES

1. Ace Equity daily prices

One large file, January 2000 onward, keyed by Ace's company code.

- Adjusted and unadjusted open, high, low, close
- Market cap, volume, traded value
- PE and P/BV
- No dividends

2. NIFTY 500 holdings workbook

Quarterly member lists from March 2012.

- Who was in the NIFTY 500 on each date
- Nifty 100 / Midcap 150 / Smallcap 250 split
- Company code ↔ ISIN ↔ symbol master, with Ace's sector

3. NSE index workbook

Daily index levels, 1996 onward. Price indices.

- NIFTY 500 (the benchmark)
- Broad indices: Midcap 150, Smallcap 250 and others
- Factor indices: Quality 50, Momentum 50, Value 50, Low Volatility 50

The paper (PDF)

Read by the model. Treated as untrusted: its text is analysed, never obeyed.

Settings in config/

- Risk-free rate 5% (fund.yaml)
- Gate B pass marks (gate_thresholds.yaml)
- Market phases (market_phases.yaml)
- Sector groups (sector_groups.yaml)

On the card, approved at Gate A: trading cost and cash rate.

BUILT ONCE BY CODE (DATA/PROCESSED/)

Price cache

The Ace file parsed into a fast table: date × stock, by ISIN. Also gives the NSE trading calendar.

Security master

One row per company: code, ISIN, symbol, sector.

Membership, point in time

Start and end date of every stock's time in the NIFTY 500, delisted names included. Version v2 fills gaps from a fund data pull; v3 uses NSE's own effective dates.

Index returns

Daily returns of the benchmark and the factor indices on the run's own days.

WRITTEN BY EACH RUN (ITS OWN FOLDER)

daily_returns.csv

Strategy and benchmark, one row per day. Everything on the Results tab starts here.

holdings.csv · trades.csv

Weights at each rebalance; every buy and sell with its price.

comparators.csv · checks · tables

The simpler alternatives, the run's own data checks, and its sensitivity tables.

WHERE EACH NUMBER ON THE PAGE COMES FROM

Tiles, returns, risk ratios, drawdown, tail risk, capture, calendar years

daily_returns.csv, compounded into complete calendar months.

Market phases

The same monthly returns, cut at the phase dates in market_phases.yaml.

Factor attribution

Monthly returns against three NSE factor indices from the index workbook.

Market-cap and sector allocation

holdings.csv, with returns from the price cache, size buckets from the member lists, sectors from the security master.

Gate B

daily_returns.csv, the run's checks, member coverage, the factor indices, the paper's date from the card.

Rule check

trades and holdings re-priced from the Ace file and compared with the approved card.

Not on file today

- NIFTY 500 total-return index: the benchmark has no dividends, so every alpha is flattered by about the dividend yield
- Benchmark sector weights
- 41 former NIFTY 500 members have no price at all
- The fund's own returns: used only as a private reference, never stored in the repo

Rules the data follows: everything is keyed by ISIN, never by symbol. A value used on a day must have been known before that day. The raw files are never changed; anything derived is written to data/processed/ or the run's folder.